

Chapter Summary

Key Terms

1.1 Basic Set Concepts

- set
- elements
- well-defined set
- empty set
- roster method
- finite set
- infinite set
- natural numbers
- integer
- set-builder notation
- cardinality of a set
- countably infinite
- equal sets
- equivalent sets

1.2 Subsets

- subset
- proper subset
- equivalent subsets
- exponential notation

1.3 Understanding Venn Diagrams

- Venn diagram
- universal set
- disjoint set
- complement of a set

1.4 Set Operations with Two Sets

- intersection of two sets
- union of two sets

Key Concepts

1.1 Basic Set Concepts

- Identify a set as being a well-defined collection of objects and differentiate between collections that are not well-defined and collections that are sets.
- Represent sets using both the roster or listing method and set builder notation which includes a description of the members of a set.
- In set theory, the following symbols are universally used:
 $\mathbb{N}$ - The set of natural numbers, which is the set of all positive counting numbers.

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

$\mathbb{Z}$ - The set of integers, which is the set of all the positive and negative counting numbers and the number zero.

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

$\mathbb{Q}$ - The set of rational numbers or fractions.

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p \text{ and } q \text{ are integers and } q \neq 0 \right\}$$

- Distinguish between finite sets, infinite sets, and the empty set to determine the size or cardinality of a set.
- Distinguish between equal sets which have exactly the same members and equivalent sets that may have different members but must have the same cardinality or size.

1.2 Subsets

- Every member of a subset of a set is also a member of the set containing it. $A \subseteq B$
- A proper subset of a set does not contain all the members of the set containing it. There is at least one member of set B that is not a member of set A . $A \subset B$
- The number of subsets of a finite set A with $n(A)$ members is equal to 2 raised to the $n(A)$ power.
- The empty set is a subset of every set and must be included when listing all the subsets of a set.
- Understand how to create and distinguish between equivalent subsets of finite and infinite sets that are not equal to the original set.

1.3 Understanding Venn Diagrams

- A Venn diagram is a graphical representation of the relationship between sets.
- In a Venn diagram, the universal set, U is the largest set under consideration and is drawn as a rectangle. All subsets of the universal set are drawn as circles within this rectangle.
- The complement of set A includes all the members of the universal set that are not in set A . A set and its complement are disjoint sets, they do not share any elements in common.
- To find the complement of set A remove all the elements of set A from the universal set U , the set that includes only the remaining elements is the complement of set A , A' .
- Determine the complement of a set using Venn diagrams, the roster method and set builder notation.

1.4 Set Operations with Two Sets

- The intersection of two sets, $A \cap B$ is the set of all elements that they have in common. Any member of A intersection B must be in both set A and set B .
- The union of two sets, $A \cup B$, is the collection of all members that are in either in set A , set B or both sets A and B combined.
- Two sets that share at least one element in common, so that they are not disjoint are represented in a Venn Diagram using two circles that overlap.
 - The region of the overlap is the set A intersection B , $A \cap B$.
 - The regions that include everything in the circle representing set A or the circle representing set B or their overlap is the set A union B , $A \cup B$.
- Apply knowledge of set union and intersection to determine cardinality and membership using Venn Diagrams, the roster method and set builder notation.

1.5 Set Operations with Three Sets

- A Venn diagram with two overlapping sets breaks the universal set up into four distinct regions. When a third overlapping set is added the Venn diagram is broken up into eight distinct regions.
- Analyze, interpret, and create Venn diagrams involving three overlapping sets.
 - Including the blood factors: A, B and Rh
 - To find unions and intersections.
 - To find cardinality of both unions and intersections.
- When performing set operations with three or more sets, the order of operations is inner most parentheses first, then find the complement of any sets, then perform any union or intersection operations that remain.
- To prove set equality using Venn diagrams the strategy is to draw a Venn diagram to represent each side of the equality or equation, then look at the resulting diagrams to see if the regions under consideration are identical. If the regions are identical the equation represents a true statement, otherwise it is not true.

Videos

1.1 Basic Set Concepts

- [Equal and Equivalent Sets \(https://openstax.org/r/Equal_and_Equivalent_Sets\)](https://openstax.org/r/Equal_and_Equivalent_Sets)

1.4 Set Operations with Two Sets

- [The Basics of Intersection of Sets, Union of Sets and Venn Diagrams \(https://openstax.org/r/operation-on-Sets\)](https://openstax.org/r/operation-on-Sets)

Formula Review

1.2 Subsets

The number of subsets of a finite set A is equal to 2 raised to the power of $n(A)$, where $n(A)$ is the number of elements

in set A : Number of Subsets of Set $A = 2^{n(A)}$.

1.4 Set Operations with Two Sets

The cardinality of A union B : $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Projects

Cardinality of Infinite Sets

In set theory, it has been shown that the set of irrational numbers has a cardinality greater than the set of natural numbers. That is, the set of irrational numbers is so large that it is uncountably infinite.

1. Perform a search with the phrase, "Who first proved that the real numbers are uncountable?"
 - a. Who first proved that the real numbers are uncountable?
 - b. What was the significance of this proof to the development of set theory and by extension other fields of mathematics?
2. Recent discoveries in the field of set theory include the solution to a 70-year-old problem previously thought to be unprovable. To learn more read [this article \(https://openstax.org/r/measure-infinities\)](https://openstax.org/r/measure-infinities):
 - a. What does it mean for two infinite sets to have the same size?
 - b. The real numbers are sometimes referred to as what?
 - c. Summarize your understanding of the problem known as the "Continuum Hypothesis."
 - d. Malliaris and Shelah's proof of this 70-year-old problem is opening up investigation in what two fields of mathematics?
3. Summarize your understanding of infinity.
 - a. Define what it means to be infinite.
 - b. Explain the difference between countable and uncountable sets.
 - c. Research the difference between a discrete set and a continuous set, then summarize your findings.

Set Notation

In arithmetic, the operation of addition is represented by the plus sign, $+$, but multiplication is represented in multiple ways, including $\cdot$, $\times$, $*$, and parentheses, such as $5(3)$. Several set operations also are written in different forms based on the preferences of the mathematician and often their publisher.

1. Search for "Set Complement" on the internet and list at least three ways to represent the complement of a set.
2. Both the Set Challenge and Venn Diagram smartphone apps highlighted in the Tech Check sections have an operation for set difference. List at least two ways to represent set difference and provide a verbal description of how to calculate the difference between two sets A and B .
3. When researching possible Venn diagram applications, the Greek letter delta, Δ appeared as a symbol for a set operator. List at least one other symbol used for this same operation.
4. Search for "List of possible set operations and their symbols." Find and select two symbols that were not presented in this chapter.

The Real Number System

The set of real numbers and their properties are studied in elementary school today, but how did the number system evolve? The idea of natural numbers or counting numbers surfaced prior to written words, as evidenced by tally marks in cave writing. Create a timeline for significant contributions to the real number system.

1. Use the following phrase to search online for information on the origins of the number zero: "History of the number zero." Then, record significant dates for the invention and common use of the number zero on your timeline.
2. Find out who is credited for discovering that the $\sqrt{2}$ is irrational and add this information to your timeline. Hint: Search for, "Who was the first to discover irrational numbers?"
3. Research Georg Cantor's contribution to the representation of real numbers as a continuum and add this to your timeline.
4. Research Ernst Zermelo's contribution to the real number system and add this to your timeline.

Chapter Review

Basic Set Concepts

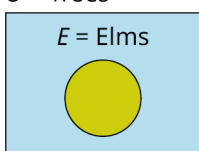
1. A _____ is a well-defined collection of distinct objects.
2. A collection of well-defined objects without any members in it is called the _____.
3. Write the set consisting of the last five letters of the English alphabet using the roster method.
4. Write the set consisting of the numbers 1 through 20 inclusive using the roster method and an ellipsis.
5. Write the set of all zebras that do not have stripes in symbolic form.
6. Write the set of negative integers using the roster method and an ellipsis.
7. Use set builder notation to write the set of all even integers.
8. Write the set of all letters in the word Mississippi and label it with a capital M .
9. Determine whether the following collection describes a well-defined set: "A group of these five types of apples: Granny Smith, Red Delicious, McIntosh, Fuji, and Jazz."
10. Determine whether the following collection describes a well-defined set: "A group of five large dogs."
11. Determine the cardinality of the set $A = \{\text{Alabama, Alaska, Arkansas, Arizona}\}$.
12. Determine whether the following set is a finite set or an infinite set: $F = \{5, 10, 15, \dots\}$.
13. Determine whether sets A and B are equal, equivalent, or neither: $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$.
14. Determine if sets A and B are equal, equivalent, or neither: $A = \{a, b, c\}$ and $B = \{c, a, b\}$.
15. Determine if sets A and B are equal, equivalent, or neither: $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$.

Subsets

16. If every member of set A is also a member of set B , then set A is a _____ of set B .
17. Determine whether set A is a subset, proper subset, or neither a subset nor proper subset of set B : $A = \{s, o, n\}$ and $B = \{s, o, n, g\}$.
18. Determine whether set A is a subset, proper subset, or neither a subset nor proper subset of set B : $A = \{s, o, n\}$ and $B = \{s, o, l\}$.
19. Determine whether set A is a subset, proper subset, or neither a subset nor proper subset of set B : $A = \{s, o, n\}$ and $B = \{o, n, s\}$.
20. List all the subsets of the set $\{\text{up, down}\}$.
21. List all the subsets of the set $\{0\}$.
22. Calculate the total number of subsets of the set $\{\text{Scooby, Velma, Daphne, Shaggy, Fred}\}$.
23. Calculate the total number of subsets of the set $\{\text{top hat, thimble, iron, shoe, battleship, cannon}\}$.
24. Find a subset of the set $\{g, r, e, a, t\}$ that is equivalent, but not equal, to $\{t, e, a\}$.
25. Find a subset of the set $\{g, r, e, a, t\}$ that is equal to $\{t, e, a\}$.
26. Find two equivalent finite subsets of the set of natural numbers, $\mathbb{N} = \{1, 2, 3, \dots\}$, with a cardinality of 4.
27. Find two equal finite subsets of the set of natural numbers, $\mathbb{N} = \{1, 2, 3, \dots\}$, with a cardinality of 3.

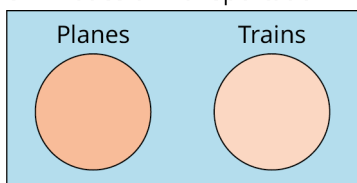
Understanding Venn Diagrams

28. Use the Venn diagram below to describe the relationship between the sets, symbolically and in words:
 $U = \text{Trees}$

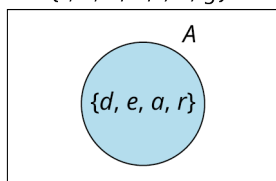


29. Use the Venn diagram below to describe the relationship between the sets, symbolically and in words:

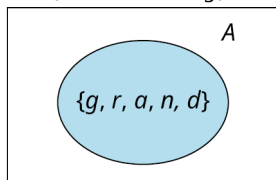
U = Modes of Transportation



30. Draw a Venn diagram to represent the relationship between the described sets: Falcons $\subset$ Raptors.
31. Draw a Venn diagram to represent the relationship between the described sets: Natural numbers $\subset$ Integers $\subset$ Real numbers.
32. The universal set is the set $U = \{s, m, i, l, e\}$. Find the complement of the set $E = \{e, l, m\}$.
33. The universal set is the set $U = \{1, 2, 3, \dots\}$. Find the complement of the set $V = \{18, 19, 20, \dots\}$.
34. Use the Venn diagram below to determine the members of the set A' .
 $U = \{r, e, a, d, i, n, g\}$



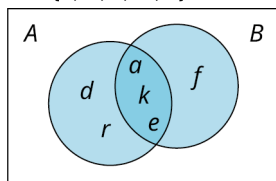
35. Use the Venn diagram below to determine the members of the set A' .
 $U = \{r, e, a, d, i, n, g\}$



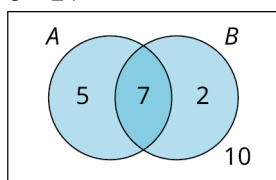
Set Operations with Two Sets

Determine the union and intersection of the sets indicated: $U = \{a, b, c, \dots, z\}$, $S = \{s, c, r, a, b, l, e\}$, $B = \{b, r, a, c, e\}$, $C = \{c, r, a, b\}$, $R = \{r, i, s, k\}$, and $Q = \{q, u, i, z\}$.

36. What is $S \cap R$?
37. What is $S \cup B$?
38. Write the set containing the elements in sets B or Q .
39. Write the set containing all the elements in both sets B and Q .
40. Find C intersection R .
41. Find C union R .
42. Find the cardinality of $C \cup R$, $n(C \cup R)$.
43. Find $n(S \text{ union } R)$.
44. Use the Venn diagram below to find $A \cap B$.
 $U = \{a, b, c, \dots, z\}$



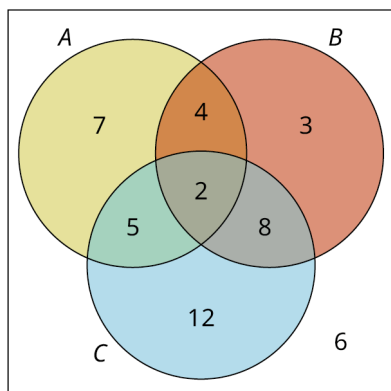
45. Use the Venn diagram below to find $n(A \cup B)$.
 $U = 24$



Set Operations with Three Sets

Use the Venn diagram below to answer the following questions.

$U = 47$



46. Find $n(A \cup C)$.
47. Find $n(B \cap C)$.
48. A food truck owner surveyed a group of 50 customers about their preferences for hamburger condiments. After tallying the responses, the owner found that 24 customers preferred ketchup, 11 preferred mayonnaise, and 31 preferred mustard. Of these customers, eight preferred ketchup and mayonnaise, one preferred mayonnaise and mustard, and 13 preferred ketchup and mustard. No customer preferred all three. The remaining customers did not select any of these three condiments. Draw a Venn diagram to represent this data.
49. Given $U = \{r, s, t, l, n, e, i, a\}$, $R = \{r, e, s, t\}$, $S = \{s, t, a, i, r\}$, and $L = \{l, i, n, e, s\}$, find $(S \cup R) \cap L'$.
50. Use Venn diagrams to prove that if $A \subset B$, then $B' \subset A'$.

Chapter Test

1. Determine whether the following collection describes a well-defined set: "A group of small tomatoes."

Classify each of the following sets as either finite or infinite.

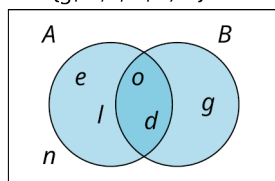
2. $\{1, 5, 9, \dots\}$
3. $\{c | c \text{ is a cat}\}$
4. $\{1, 2, 3, \dots, 1000\}$
5. $\{s, m, i, l, e\}$
6. $\{m \in \mathbb{N} | m = n^2 \text{ where } n \text{ is a natural number}\}$

Use the sets provided to answer the following questions: $U = \{31, 32, 33, \dots, 50\}$, $A = \{35, 38, 41, 44, 47, 50\}$, $B = \{32, 36, 40, 44, 48\}$, and $C = \{31, 32, 41, 42, 48, 50\}$.

7. Find A or B .
8. Find B and C .
9. Determine if set A is equivalent to, equal to, or neither equal nor equivalent to set C . Justify your answer.
10. Find $n(A \cup C)$.
11. Find $A \cap (B \cap C)$.
12. Find $(A \cup B)' \cap C$.
13. Find $(A \cap B') \cup C$.

Use the Venn diagram below to answer the following questions.

$U = \{g, o, l, d, e, n\}$

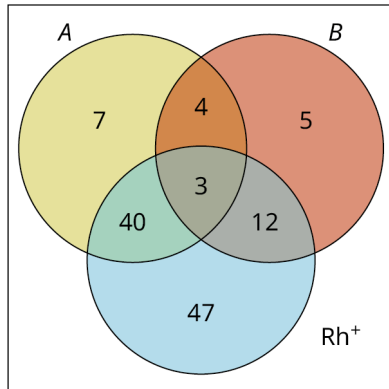


14. Find B' .
15. Find $A \cup B$.
16. Find $A \cap B'$.
17. Draw a Venn diagram to represent the relationship between the two sets: "All flowers are plants."

For the following questions, use the Venn diagram showing the blood types of all donors at a recent mobile blood

drive.

Donors = 128



18. Find the number of donors who were O^- ; that is, find $n((A \cup B \cup Rh^+)')$.
19. Find the number of donors who were A^+ or B^+ or AB^+ .
20. Use Venn diagrams to prove that if $A \subset B$, then $A \cap B = A$.

